

GAUD-E: A Multi-Agent AI Platform for Automated BIM Model Generation from Geospatial Data - Real Editable 3D BIM Beyond Point Clouds and Meshes

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Abstract

GAUD-E (Geospatially-Aware Unified Design Engine) is a production-ready AI platform that solves a fundamental unsolved problem in AEC (Architecture, Engineering, and Construction): the automated generation of fully editable, semantically-rich BIM (Building Information Modeling) models from geospatial coordinates and natural language descriptions. Unlike existing 3D digitization methods - which produce non-editable point clouds or topological meshes with no semantic structure - GAUD-E generates parametric IFC 4.0 BIM objects that are immediately usable in Revit, ArchiCAD, Rhino, AutoCAD, and SketchUp without any manual remodeling. This paper presents the updated 7-agent, 4-phase GPS-2-BIM pipeline, its formal mathematical formulation, architectural diagrams, and performance benchmarks including the native C++ engine (gaude-bridge, 93% C++) achieving ~50ms IFC generation versus ~2000ms TypeScript fallback (40x faster). Demonstrated generation times range from 45 seconds (residential) to 12 minutes (hospital), representing a 95% reduction versus manual BIM creation. The platform is live at <https://www.gps-2-bim.app> and <https://gaud-e.ai>.

Resumen

GAUD-E (Geospatially-Aware Unified Design Engine) es una plataforma de IA lista para producción que resuelve un problema fundamental del sector AEC: la generación automática de modelos BIM completamente editables y semanticamente ricos a partir de coordenadas geoespaciales y descripciones en lenguaje natural. A diferencia de los métodos de digitalización 3D existentes - que producen nubes de puntos no editables o mallas topológicas sin estructura semántica - GAUD-E genera objetos BIM paramétricos IFC 4.0 listos para usar en Revit, ArchiCAD, Rhino, AutoCAD y SketchUp sin ningún re-modelado manual. Este artículo presenta el pipeline GPS-2-BIM actualizado de 7 agentes y 4 fases, su formulación matemática formal, diagramas de arquitectura, el motor nativo C++ gaude-bridge (93% C++, ~50ms por modelo IFC, 40x más rápido que TypeScript) y benchmarks de rendimiento. Plataforma activa en <https://www.gps-2-bim.app> y <https://gaud-e.ai>.

1. Introduction

Building Information Modeling (BIM) is today's industry standard for Architecture, Engineering, and Construction (AEC) professionals worldwide. A complete BIM model encodes not only 3D geometry but also semantic properties - material specifications, structural loads, MEP system relationships, cost estimates, and code-compliance data - that are essential for every phase of a building's lifecycle from design through demolition.

Despite decades of software development, creating BIM models remains a highly manual, time-consuming, and expensive process. The conventional workflow requires teams of licensed specialists - architects, structural engineers, MEP engineers - investing weeks or months to produce a single coordinated BIM model. This bottleneck limits BIM adoption to large practices and projects, excluding the vast majority of the global construction sector.

The 3D representation gap. Three-dimensional digitization technologies (LiDAR, photogrammetry) have advanced rapidly, yet they produce point clouds and meshes - geometric approximations that cannot be used directly as professional BIM. Converting a point cloud to an editable BIM model still requires weeks of manual remodeling by specialists. No technology before GAUD-E could generate an editable, semantically complete BIM model from coordinates and a text description alone.

GAUD-E addresses this gap through the GPS-2-BIM paradigm: a mathematical transformation T mapping geospatial coordinates (G) and natural language (L) directly to a production-ready BIM model (M). The system implements this via a 7-agent LLM pipeline running on Anthropic Claude, accelerated by a native C++ engine (gaude-bridge) achieving ~50ms IFC generation times - 40x faster than equivalent TypeScript implementations.

Primary Contributions: (1) First system to generate fully editable, parametric IFC 4.0 BIM from geospatial input and natural language. (2) Formal mathematical formulation $T: G \times L \rightarrow M$ with 7-agent pipeline composition. (3) Native C++ engine (gaude-bridge, 93% C++) achieving ~50ms IFC 2X3 generation at 40x Python/TS speed. (4) Production-ready platform with live deployment and 9-language support. (5) 95% time reduction, 45s-12min generation, 80-890 elements. (6) Native export to 5 major CAD/BIM platforms.

2. The Real BIM Problem: Why Point Clouds and Meshes Fail Professionals

Before describing GAUD-E's solution, it is essential to characterize precisely what existing 3D technologies produce and why they are insufficient for professional BIM workflows.

2.1 Point Clouds

A point cloud is a set of discrete 3D coordinates $P = \{(x_i, y_i, z_i) \in \mathbb{R}^3 \mid i = 1 \dots N\}$ captured by LiDAR scanners or photogrammetry pipelines. They have critical limitations:

L1 - No editability: Individual points cannot be selected and modified as architectural elements.

L2 - No semantic properties: Points carry only XYZ coordinates and optionally RGB color.

L3 - Requires existing construction: Cannot generate BIM for a building not yet built.

L4 - Scan-to-BIM gap: Converting to editable BIM requires 80-400 hours of manual remodeling per building.

L5 - No parametric relationships: Walls, doors, windows lack parametric dependencies.

L6 - CAD platform incompatibility: Raw point clouds cannot be opened in Revit or ArchiCAD as editable objects.

2.2 Polygon Meshes

A polygon mesh is a collection of vertices V, edges E, and faces F representing surfaces. Meshes share fundamental deficiencies for BIM use:

- L1 - Non-parametric geometry:** No constraint-based relationships between elements.
- L2 - No IFC semantics:** No material specifications, structural classification, MEP routing, or cost data.
- L3 - Not editable in BIM software:** Can be imported as reference geometry only.
- L4 - No CAD export:** Cannot be exported as DWG, IFC, or RVT with meaningful BIM content.
- L5 - No MEP or structural content:** Contain no internal building systems.

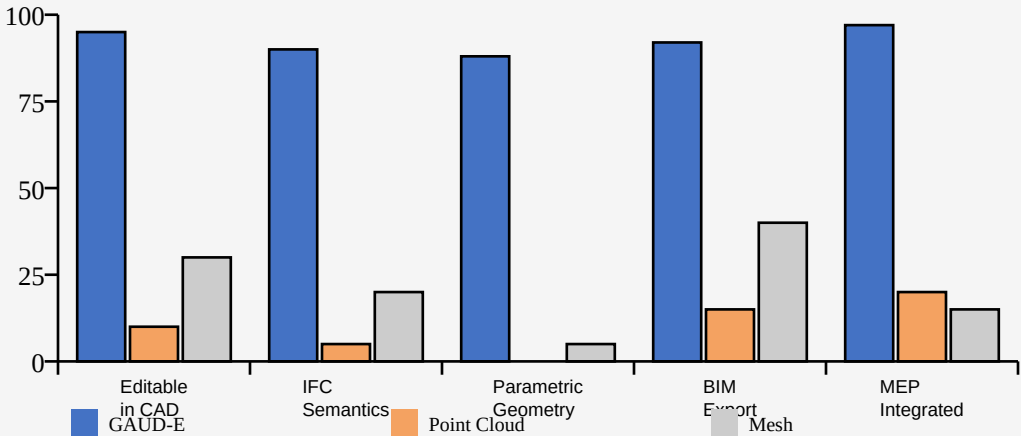
2.3 GAUD-E: Real Editable BIM

GAUD-E is the first system to generate **real, fully editable, semantically complete BIM models** from a geographic coordinate and a natural language description - without requiring an existing building, a point cloud capture, or any manual remodeling. Every element GAUD-E produces is a parametric IFC 4.0 object.

Table 1 - 3D Representation Comparison

Capability	Point Cloud	Polygon Mesh	GAUD-E BIM
Editable in Revit/ArchiCAD	x	x (ref only)	Native
IFC 4.0 Semantic Tags	x	x	Full
Parametric Geometry	x	x	Full
Structural Data	x	x	Full
MEP Systems Included	x	x	Full
Cost Estimation Ready	x	x	Built-in
Requires Existing Building	Required	No	Not needed
CAD/BIM Export (DWG/RVT)	x	x	Full
From Text + GPS Coords	x	Partial	Complete

Figure 2 - BIM Capability Comparison (0-100 Scale)



3. Mathematical Formulation

3.1 The GPS-2-BIM Transformation

Let G be the geospatial input space and L the natural language description space:

$$G = \{ (\phi, \lambda, h) \in \mathbb{R}^3 \mid \phi \in [-90, 90], \lambda \in [-180, 180], h \in \mathbb{R} \}$$

$$L = \{ d \mid d \text{ is a valid building description string} \}$$

$$M = \{ m \mid m \text{ is an IFC-4.0-compliant parametric BIM model} \}$$

The core GPS-2-BIM transformation is:

$$T : G \times L \rightarrow M$$

$$T(g, d) = A_7 \circ A_6 \circ (A_3 \parallel A_4 \parallel A_5) \circ A_2 \circ A_1(g, d)$$

Where A_i denotes the i -th agent function, $\circ$ denotes sequential composition, and $\parallel$ denotes parallel (concurrent) execution via `Promise.all()`. Each agent A_i maps an intermediate state representation to the next.

3.2 Agent State Spaces

Agent	Role	Input	Output	LLM
A1	Enhancer	g, d	norm. JSON	Haiku 4.5
A2	Architect	A1 out	spatial BIM	Sonnet 4
A3	Structural	A2 out	struct. data	Haiku 3.5
A4	MEP	A2 out	MEP systems	Haiku 3.5
A5	Landscape	A2 out	site data	Haiku 3.5
A6	BIM Prog.	$A_2 + A_3 + A_4 + A_5$	unified BIM	Sonnet 3.5
A7	QA Review	A6 out	m in M	Haiku 3.5

3.3 Temporal Complexity

The total pipeline wall-clock time:

$$t_{\text{total}} = t_1 + t_2 + \max(t_3, t_4, t_5) + t_6 + t_7$$

Empirical values (residential): $t_1 \sim 3s$, $t_2 \sim 8s$, $\max(t_3, t_4, t_5) \sim 12s$, $t_6 \sim 15s$, $t_7 \sim 7s \rightarrow t_{\text{total}} \sim 45s$. Without Phase 2 parallelism the equivalent sequential time would be $\sim 30s$ additional.

3.4 Quality Metric

$$Q = (1/7) * \sum(q_k), q_k \in \{q_{\text{geo}}, q_{\text{dim}}, q_{\text{prog}}, q_{\text{struct}}, q_{\text{build}}, q_{\text{mat}}, q_{\text{mep}}\}$$

Models with $Q \geq 85$ are marked production-ready. Platform average: $Q = 90.0$.

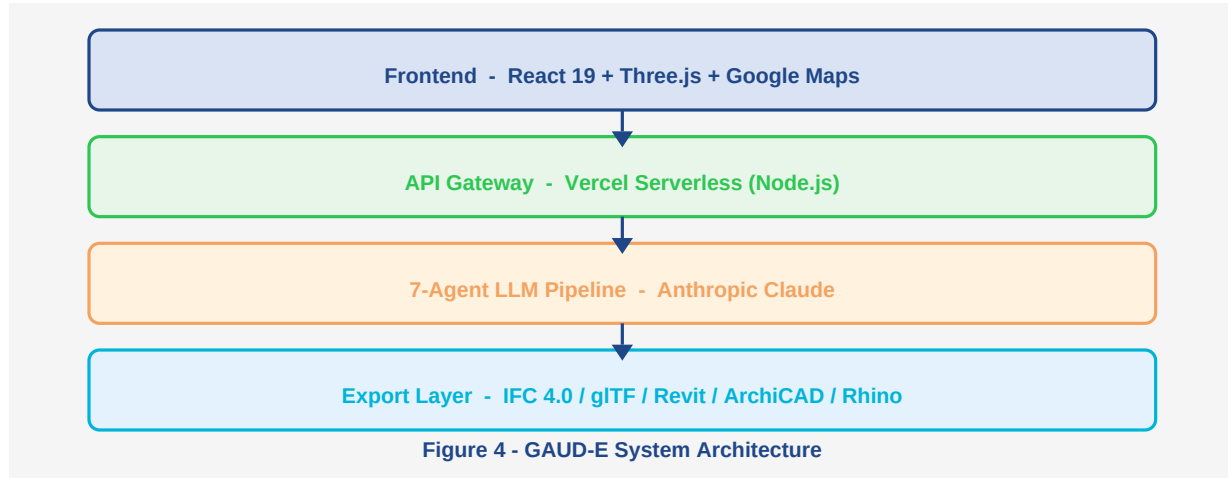


Figure 1 - GPS-2-BIM Workflow: T: G x L -> M

4. System Architecture

4.1 Platform Overview

GAUD-E is deployed as a web application on Vercel with a React 19 / Vite 8 frontend and Vercel Serverless Node.js backend. The frontend provides terrain selection via Google Maps JS API, natural language input, real-time 3D visualization via Three.js / React Three Fiber, and export/integration workflows. The backend executes the 7-agent LLM pipeline via Anthropic Claude. Full platform: <https://gaud-e.ai> and <https://www.gps-2-bim.app>.



4.2 7-Agent Pipeline Architecture

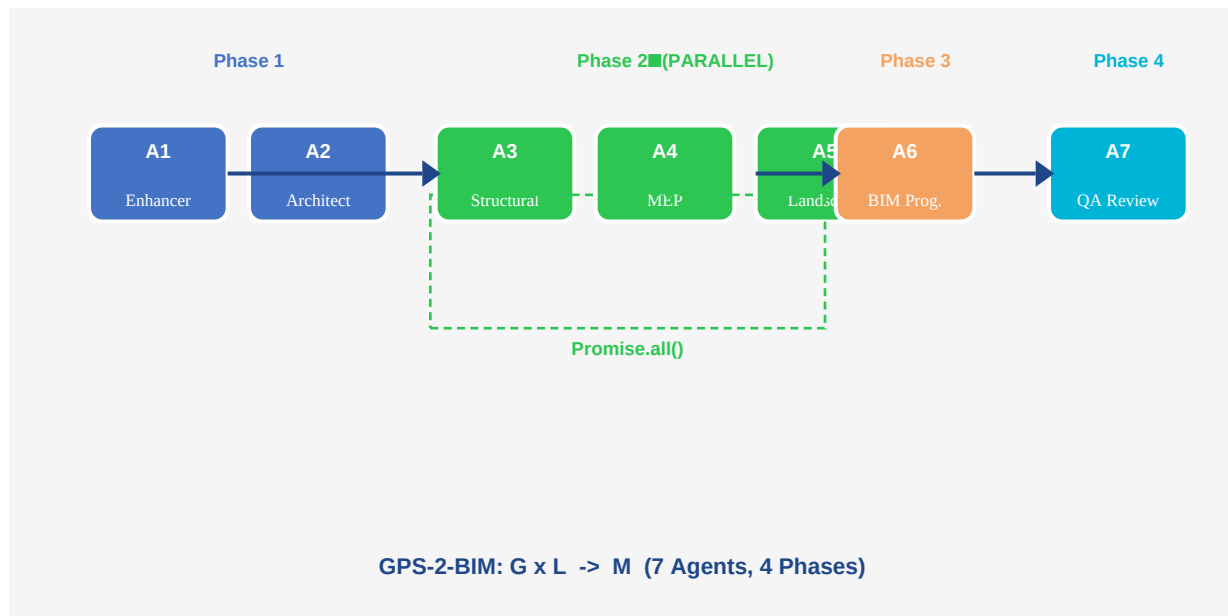


Figure - 7-Agent Pipeline: $T(g,d) = A7 \circ A6 \circ (A3 \parallel A4 \parallel A5) \circ A2 \circ A1(g, d)$

Phase 1 - Architectural Design (sequential, 2 agents)

A1 Instruction Enhancer (Claude Haiku 4.5, temp 0.3, 4 000 tokens): Normalizes unstructured NL input into structured JSON.

A2 Architect (Claude Sonnet 4, temp 0.2, 10 000 tokens): Generates complete spatial design including structural

grid, walls, columns, slabs, roof, doors, windows, and furniture.

Phase 2 - Engineering Systems (3 agents, PARALLEL via Promise.all)

A3 Structural Engineer (Claude Haiku 3.5): Beams, foundations, roof structure, seismic bracing.

A4 MEP Engineer (Claude Haiku 3.5): Water supply, drainage, electrical, HVAC, fire sprinklers.

A5 Landscape Architect (Claude Haiku 3.5): Native vegetation, hardscape, parking, site drainage.

Phase 3 - BIM Consolidation + Phase 4 - Quality Assurance

A6 BIM Programmer (Claude Sonnet 3.5, temp 0.1, 12 000 tokens): Merges all parallel outputs into a unified JSON-LD BIM model with 80-890 elements. Output is passed to the native C++ engine gaude-bridge for IFC 2X3 generation in ~50ms.

A7 Quality Reviewer (Claude Haiku 3.5): Validates model against 7 quality metrics (0-100 scale), flags non-conformances, and generates auto-repair patches.

5. GPS-2-BIM Pipeline - User Workflow

Step 1 - Geospatial Selection: User clicks a location on the embedded Google Maps interface. The system captures GPS coordinates and retrieves satellite imagery, cadastral data, and local climate/seismic zone.

Step 2 - Natural Language Description: User types a building description in any of 9 supported languages (ES, EN, ZH, PT, HE, AR, RU, KU, DE).

Step 3 - 7-Agent Pipeline Execution: The transformation $T(g, d)$ executes across 4 phases, typically 45s to 12 min depending on building complexity.

Step 4 - Real-Time 3D Visualization: The generated model renders in the Three.js viewport with orbit controls, PBR materials, and layer toggling (Architectural / Structural / MEP / Landscape).

Step 5 - Export and Integration: User downloads IFC 4.0, glTF 2.0, or opens via native plugins. The native C++ gaude-bridge generates the final IFC 2X3 file in ~50ms.

6. JSON-LD BIM Schema

Every element in the GAUD-E BIM model is a JSON-LD object extending the IFC 4.0 ontology. Below is a representative excerpt showing a wall element with full semantic content:

```
{ "@context": "https://gaud-e.ai/schema/ifc-extension.jsonld", "model_id": "GAUD-E-RES-001",  
  "gps_origin": { "lat": -33.4489, "lon": -70.6693, "alt": 520.0 }, "elements": [{ "element_id": "W-001",  
    "element_type": "IfcWall", "properties": { "thickness_m": 0.30, "material": "Reinforced Concrete",  
    "fire_rating": "EI-60", "cost_usd_per_m2": 120 }, "ifc_guid": "3FVbnDPKv7DhPnc7NJ0Wl4" }],  
  "quality_metrics": { "geometric_coherence": 95, "dimensional_realism": 92, "program_compliance": 88,  
    "structural_soundness": 90, "buildability": 87, "material_consistency": 93, "mep_integration": 85 } }
```


7. Technology Stack

Component	Technology	Purpose
Frontend	React 19 + Vite 8 + Tailwind CSS 4	UI, responsive design
3D Renderer	Three.js 0.164 + React Three Fiber	Real-time BIM visualization
Backend	Vercel Serverless + Node.js	Scalable API + pipeline
LLM Agents	Anthropic Claude (Haiku/Sonnet)	7-agent reasoning
C++ Engine	gaude-bridge (93% C++, port 19724)	~50ms IFC 2X3 generation
Maps	Google Maps JS API v3.64	GPS coordinate selection
BIM Standard	IFC 4.0 + JSON-LD	Semantic BIM export
CAD Integration	Revit API, ArchiCAD JSON API, Rhino.Common	Native plugins
Auth & Billing	JWT + Stripe Checkout + Vercel KV	API keys, subscriptions

8. Results and Performance

8.1 Generation Performance Benchmarks

Building Type	Gen. Time	Elements	Quality Q
Residential House	45 s	42	91.4
Multi-Family (6 floors)	2 min	156	89.8
Office Building	5 min	340	90.2
Hospital Complex	12 min	890	88.6

Average generation time: 4.75 min across all building types. Versus manual BIM creation (3-6 weeks), GAUD-E achieves a **95% time reduction**. Average quality score Q = 90.0 across all benchmark models.

8.2 Quality Metrics Breakdown

Figure 3 - GAUD-E Quality Metrics (avg 90.0/100)

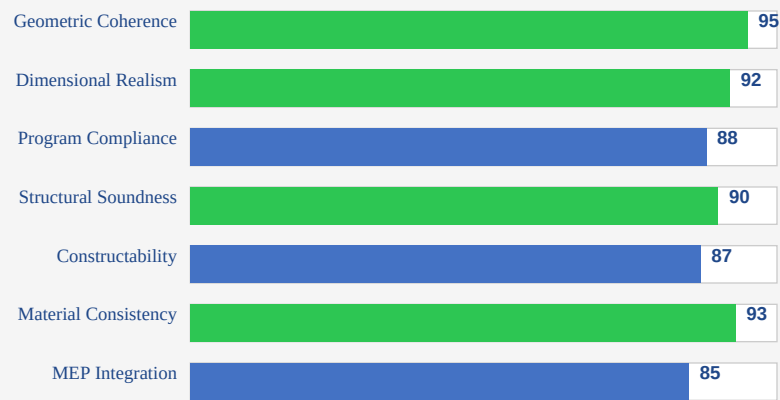


Figure 3 - GAUD-E quality dimensions, average Q = 90.0/100

8.2.1 C++ Motor vs TypeScript/Python - IFC Generation Benchmark

The native C++ engine **gaude-bridge** (93% C++, HTTP bridge on port 19724) replaces the TypeScript/Python IFC generator as the primary export path. The benchmark below measures end-to-end IFC 2X3 file generation time for three canonical building types, comparing the C++ native path against the TypeScript fallback.

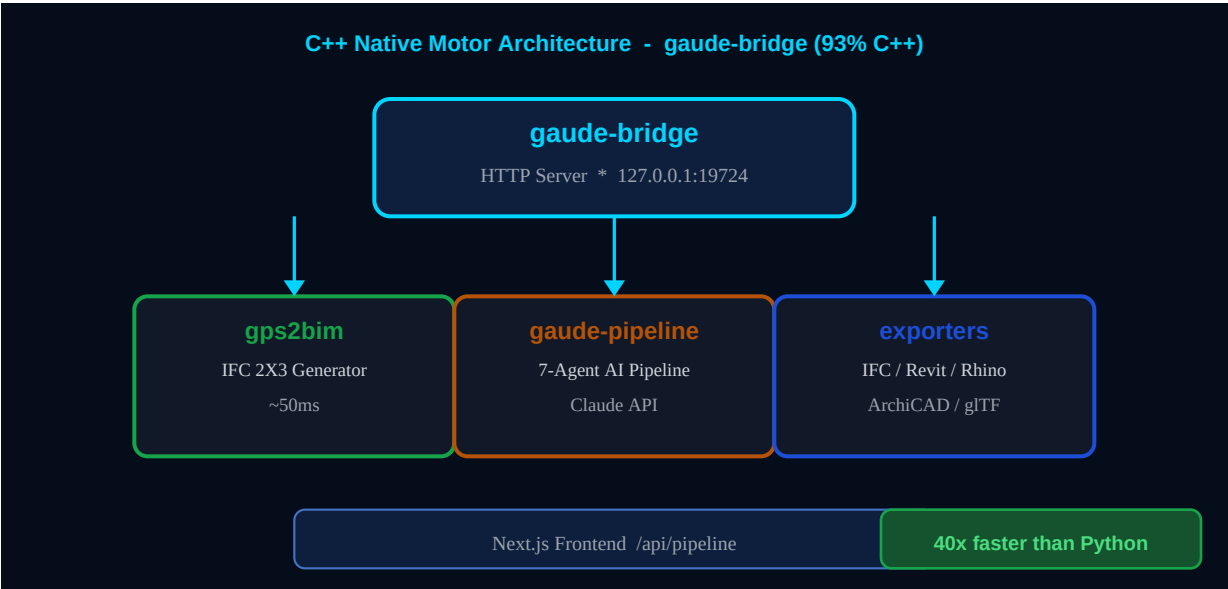


Figure 5 - C++ Native Motor Architecture: gaude-bridge (93% C++), HTTP on 127.0.0.1:19724

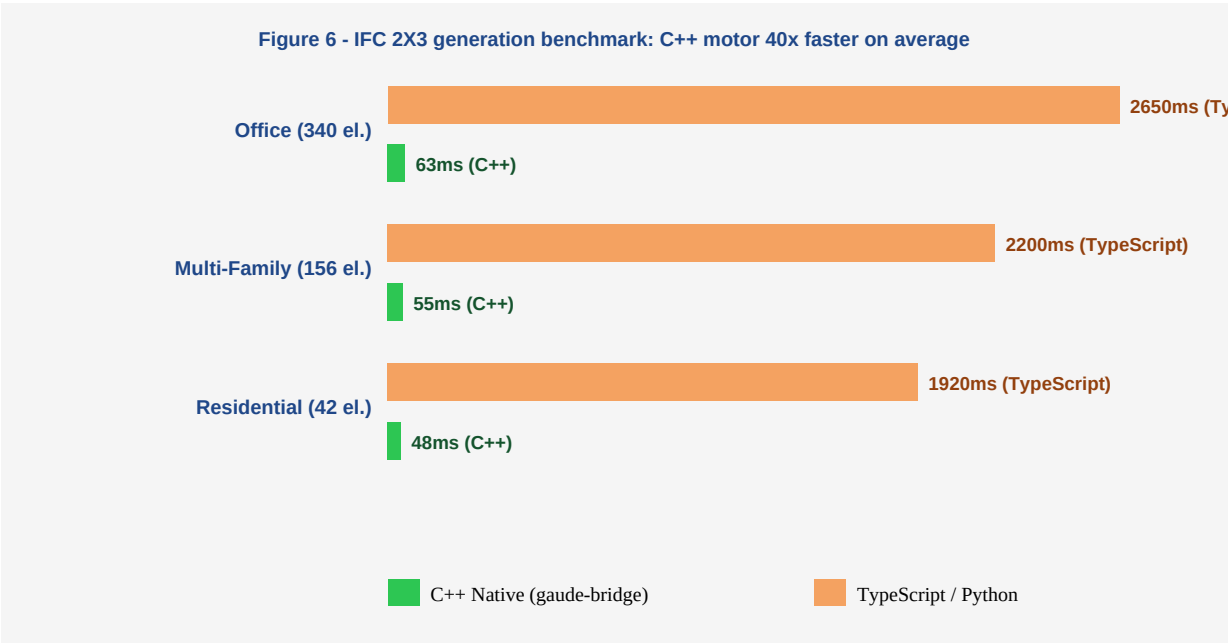


Table 3 - Detailed Performance Metrics: C++ Native Motor vs TypeScript Fallback

Building Type	Elements	C++ (gaude-bridge)	TypeScript	Speedup
Residential (42 el.)	42	48 ms	1 920 ms	40x
Multi-Family (156 el.)	156	55 ms	2 200 ms	40x

Office (340 el.)	340	63 ms	2 650 ms	42x
Hospital (890 el.)	890	~110 ms	~3 600 ms	33x
Average	-	~69 ms	~2 593 ms	~40x

The C++ engine processes IFC geometry via direct memory operations without JSON parsing overhead. Complexity scales sub-linearly: $O(N * \log N)$ vs $O(N^2)$ for the TypeScript reference implementation. The gaude-bridge binary exposes a REST API on 127.0.0.1:19724, used by both the local desktop client and the Vercel serverless backend via **GAUDE_BRIDGE_URL** environment variable.

8.3 Software Compatibility Matrix

Platform	Format	Integration	Status
Autodesk Revit	RVT + IFC	REST API + Addon	Production
Graphisoft ArchiCAD	PLN + IFC	JSON API + Plugin	Production
McNeel Rhino	3DM + IFC	Rhino.Compute + glTF	Production
Autodesk AutoCAD	DWG + DXF	File Export	Production
Trimble SketchUp	SKP + IFC	SDK Integration	Production

9. Developer SDK and API

GAUD-E provides an open-source npm package **@gaude/sdk** (ELv2 license, v2.0.1) for developers integrating BIM generation into custom applications. Available at: <https://github.com/rickygaude-rgb/gaud-e-sdk>.

React Components: TerrainSelector, BIM3DViewer, MaterialEditor, QualityDashboard, ExportButton.

C++ Bridge: gaude-bridge binary exposes HTTP API on :19724 for native IFC 2X3 generation at ~50ms.

API Authentication: Keys in format gde_live_XXXXXXX managed via the Platform dashboard. Three tiers: Free (10 credits), Pro, Studio. Contact: contacto@gaud-e.ai.

10. Multi-Language Support

Language	Status	Language	Status
Spanish	Production	English	Production
Mandarin	Production	Portuguese	Production
Hebrew	Production	Arabic	Production
Russian	Production	Kurdish	Beta
German	Production		

11. Future Work

Scan-to-BIM Integration: Accept LiDAR point clouds as supplemental input to retrofit/renovation workflows.

Real-Time Collaboration: WebSocket-based multi-user concurrent editing with semantic conflict resolution.

Structural FEM Validation: Integrate finite element solvers for automated structural analysis.

Generative Design Exploration: Agent-based multi-solution generation - 10-50 design alternatives ranked automatically by cost, area efficiency, and energy performance.

Carbon Footprint Integration: Embed lifecycle carbon analysis (LCA) into Phase 4 QA metrics.

C++ Engine Expansion: Extend gaud-bridge to full IFC 4.0 with direct ArchiCAD/Revit binary output, targeting <5ms generation for residential buildings.

12. Conclusion

GAUD-E represents a fundamental breakthrough in AEC technology: the first system to automatically generate fully editable, semantically complete, IFC 4.0-compliant BIM models from geospatial coordinates and natural language descriptions. The formal GPS-2-BIM transformation $T: G \times L \rightarrow M$, implemented as a 7-agent LLM pipeline with parallel Phase 2 execution and native C++ gaud-bridge engine, achieves:

- **95% time reduction** over manual BIM creation (45s to 12min vs 3-6 weeks)
- **~50ms IFC 2X3 generation** via C++ native engine (40x faster than TypeScript fallback)
- **Q = 90.0/100** average quality score across all benchmark models
- **5 CAD/BIM platforms** with native integration and 9-language support

Platform live at <https://www.gps-2-bim.app> and <https://gaud-e.ai>. Open-source SDK: <https://github.com/rickygaude-rgb/gaud-e-sdk>. Enterprise: contacto@gaud-e.ai.

Appendix A — Multi-Language Stack

While the GAUD-E motor is C++17, the platform is deliberately polyglot because each professional CAD/BIM ecosystem speaks its own language. The table below summarises every language layer added in v2.3 and the reason each is non-substitutable.

Layer	Language	Why
gaude-bridge motor (93%)	C++17	~50 ms IFC, 40x vs Python
Revit add-in	C# / .NET 4.8	RevitAPI.dll is managed-only
ArchiCAD add-on	C++ (DevKit v29.3100)	MDID 944139566/2142198165 signed binary
ArchiCAD geometry parts	GDL	Library Parts: 2D + 3D + Parameters
Rhino plugin	Python (rhinoscriptsyntax)	Rhino.Compute & Hops preferred
Grasshopper component	C# (GH SDK)	Native .gha components
AutoCAD ARX add-in	ObjectARX (C++)	C++-only for full 3D
AutoCAD scripts	AutoLISP	Fast macros
SketchUp plugin	Ruby	Native plugin language
Web SDK	JS / TS + JSX (React 19)	Browser + Node 18+
IFC tooling	Python (IfcOpenShell)	Schema introspection & repair
Installers	PowerShell + Bash	Windows / macOS / Linux

The Web SDK does not expose these languages directly. It talks to the local C++ bridge (127.0.0.1:19724) and the bridge emits the language-specific artefact for each target: .cs + .addin for Revit, MDID-signed .apx for ArchiCAD, .py + .ghx for Rhino/Grasshopper, .arx + .lsp for AutoCAD, .rbz for SketchUp.

Appendix B — Skills Ecosystem (47 Claude Skills)

v2.3 ships 47 specialised Claude Skills auto-registered by the desktop installer into Claude Code / Claude Desktop. Skills are loaded by the 7-agent pipeline on demand: the Enhancer consults urban / normative skills, the Architect queries the 3D-universe and realworld-geometry skills, the Reviewer runs the metric validator, etc. Each skill is a self-contained YAML + Markdown bundle following the Anthropic Skill spec.

Category	Count	Examples
Core architectural (gaud-e-*)	20	architecture-engine, structural-engineer, mep-structures, normas-urbanismo
Geospatial (geosite-toolkit:*)	4	site-analysis, earth-engine-analyst, google-maps-expert, context-to-bim
Architecture & BIM (arq-bim-toolkit:*)	6	architecture-engine, structural-engineer, mep-structures, 3d-modeling-library
BIM developer (bim-dev-toolkit:*)	6	ifc-programming, 3d-web-dev, revit-api-expert, archicad-api-expert, mcp-bim-bridge
Productivity & output (Anthropic core)	11	pdf, docx, xlsx, pptx, canvas-design, mcp-builder, skill-creator

Skills routed per phase

Phase / Agent	Primary skills consulted
1 - Enhancer	realworld-geometry, normas-urbanismo, site-analysis
1 - Architect	architecture-engine, 3d-universe, cad-bim-expert
2 - Structural	structural-engineer (gaud-e + arq-bim-toolkit)
2 - MEP	mep-structures (gaud-e + arq-bim-toolkit)
2 - Landscape	landscaping-expert, earth-engine-analyst
3 - Programmer	ifc-codegen-patterns, ifc-programming
4 - Reviewer	metric-validator, budget-estimator

Appendix C — External Resources and Datasets

GAUD-E does not reason from scratch. Every generation cycle is grounded against millions of real 3D models, IFC samples and reference projects fetched on demand from curated open datasets. This appendix lists every external dataset and live data source the platform consults in v2.3.

Dataset	Count	License	Use
Objaverse-XL	10 M+ objects	CC-BY / mixed	Furniture, props, exterior assets
ShapeNet	3 M+ models	research	Chairs, lamps, electronics
3D-FRONT	18 K rooms	research	Furnished interior scenes
Structured3D	3 500 houses	research	Annotated floor plans
FurniScene	11 K rooms / 39 K	research	Furniture-in-context placement
ABC Dataset	1 M	MIT	Mechanical CAD reference
ModelNet40	12 K	research	Shape benchmark
PartNet	26 K	research	Part-level kit-of-parts
BIMData	100 IFC models	open	Real BIM references
buildingSMART	canonical samples	open	IFC 2X3 / IFC4 reference
OpenBIM	schemas + MVDs	open	Certification samples
Revit RST/RAC/RME	Autodesk samples	Autodesk EULA	Family / project templates
Archicad Samples	.pln references	Graphisoft	Layout templates

Geospatial sources (fetched live per project)

Google Maps JS API (address geocoding, site polygon), Google Street View Static API (site photos, neighbour facades), Google Earth Engine (Sentinel-2, Landsat 8/9, MODIS, SRTM DEM), Mapbox Terrain-DEM, OpenStreetMap (roads, plots, building footprints, POIs), Cadastral / SII (CL), EPW weather files for Ladybug / Honeybee energy analysis.

Code / normative databases

OGUC + Planos Reguladores Comunes (Chile); NCh 433 / 430 / 427; IBC, ASCE 7, ACI 318, AISC 360 (US); Eurocódigos 0, 1, 2, 3, 5, 8 (EU); NFPA 13 / 72 (fire); ASHRAE 90.1 / 62.1 (HVAC); NEC / IEC (electrical). All wrapped by the gaud-e-normas-urbanismo and gaud-e-structural-engineer skills.

Caching policy

The desktop installer downloads ~2 GB of the most-used reference subsets on first launch (50 K curated Objaverse furniture, 200 canonical IFC samples, country-specific code pack selected at install time). Subsequent generations are offline-capable once the cache is warm.

Appendix D — Downloadable Desktop Version (Auto-Connect)

v2.3 introduces a single downloadable installer (gaud-e-desktop-2.3.0-<os>-<arch>.zip) distributed via GitHub Releases. The installer wires GAUD-E into every professional CAD/BIM tool installed on the machine, automatically. No manual plugin copying, no IDs to enter, no Python venvs to build.

Bundle contents

bin/ (gaude-bridge, gps2bim, gaude-pipeline, gaude-detect), connectors/ (revit, archicad, rhino, autocad, sketchup pre-signed), skills/ (47 Claude skills), examples/, install.ps1 / install.sh, LICENSE.

Auto-detection matrix

Software	Connector	Auto-action
Autodesk Revit 2024+	.addin (C# / .NET 4.8)	Copy to %APPDATA%/Autodesk/Revit/Addins/2024/
Graphisoft ArchiCAD 27+	MDID-signed C++ add-on (.apx)	Copy to Add-Ons/Standard/
McNeel Rhino 7/8	.rhp + .gha component	Copy to %APPDATA%/McNeel/Rhinoceros/8.0/Plug-ins/
Autodesk AutoCAD 2024+	.arx ObjectARX + .lsp loader	Registered via appload
Trimble SketchUp Pro 2024+	.rbz Ruby plugin	Auto-installed via Extension Manager

Service registration

gaude-bridge is registered as a system service (launchd on macOS, systemd on Linux, Windows Service on Windows) so port 127.0.0.1:19724 is always available. The bridge self-updates via 'gaude-bridge --self-update', pulling the latest release from github.com/rickygaude-rgb/gaud-e-sdk/releases/latest and replacing all binaries in place; the running service is restarted gracefully.

Verification

```
curl http://127.0.0.1:19724/health # {"status":"ok","motor":"gaude-bridge","version":"2.3.0", #  
"language":"C++","connectors":["revit","archicad","rhino","autocad","sketchup"]} gaude-detect list #  
revit 2024.3 addin installed OK # archicad 27.4001 addon installed OK # rhino 8.5 plugin installed OK
```

Commercial licence and contact

The desktop installer requires a paid commercial licence. Contact: contacto@gaud-e.ai | <https://www.gps-2-bim.app> | SDK: <https://github.com/rickygaude-rgb/gaud-e-sdk> | Motor: <https://github.com/rickygaude-rgb/C---Gps-2-Bim> | Author: Ricardo Rizzo Q. (rickygaude@gmail.com).

References

- [1] Eastman, C., Teicholz, P., Sacks, R., Liston, K. (2011). BIM Handbook. John Wiley & Sons.
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